# Comprehensive Test Report

Regression Tests • Benchmarks • Physics Diagnostics

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**VegasAfterglow 2.0.2.dev76 | Python 3.12.9**

Commit: 2bafcaf1 | Platform: Darwin arm64

CPU: Apple M2

*Powered by Claude Code*

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# Regression Tests

Power-law scaling verification against theoretical predictions

# Regression Test Report Guide

This document describes the regression validation framework for VegasAfterglow, which verifies that simulation outputs reproduce the expected power-law scaling relations from GRB afterglow theory.

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## 1. Test Categories

The regression suite validates physical quantities against analytical predictions derived from standard synchrotron afterglow theory. Tests cover both forward and reverse shocks:

**Forward shock:**

1. **Shock dynamics**: Lorentz factor, radius, magnetic field, particle number 2. **Characteristic frequencies**: $\nu_m$ (injection), $\nu_c$ (cooling), $\nu_M$ (maximum) 3. **Spectral shapes**: Power-law indices in different frequency regimes

**Reverse shock:**

4. **Thin-shell dynamics**: Shock dynamics during and after reverse shock crossing (short engine duration) 5. **Thick-shell dynamics**: Shock dynamics during and after reverse shock crossing (long engine duration) 6. **Thin-shell frequencies**: Characteristic frequency evolution during crossing phase 7. **Thick-shell frequencies**: Characteristic frequency evolution during crossing phase

All regression tests use a model resolution of `(0.3, 2, 15)` to ensure well-resolved evolution for accurate power-law fitting.

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## 2. Evolutionary Phases

### 2.1 Forward Shock

The blast wave passes through distinct dynamical phases as it decelerates. Each phase exhibits characteristic power-law behavior that serves as a validation target.

| Phase | Physics |
|---|---|
| Coasting | Free expansion, $\Gamma \approx$ const |
| Blandford-McKee | Self-similar deceleration |
| Sedov-Taylor | Non-relativistic, u < 0.1 |

### 2.2 Reverse Shock (Thin Shell)

Short engine duration. The reverse shock crosses the ejecta quickly, then evolves as a decaying blast wave.

| Phase | Physics |
|---|---|
| Crossing | Reverse shock traversing ejecta |
| Post-crossing | Post-crossing deceleration |
| Sedov-Taylor | Non-relativistic |

### 2.3 Reverse Shock (Thick Shell)

Long engine duration. The reverse shock crosses while the engine is still active, producing different scaling.

| Phase | Physics |
|---|---|
| Crossing | Reverse shock traversing during engine activity |
| Post-crossing | Post-crossing deceleration |
| Sedov-Taylor | Non-relativistic |

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## 3. Expected Scaling Relations

### 3.1 Forward Shock Dynamics

Physical quantities at the shock front scale as power laws with observer time: $Q \propto t^{\alpha}$. The exponents depend on the external medium density profile.

**ISM (constant density n)**

| Phase | u | r | B | $N_p$ |
|---|---|---|---|---|
| Coasting | 0 | 1 | 0 | 3 |
| Blandford-McKee | -3/8 | 1/4 | -3/8 | 3/4 |
| Sedov-Taylor | -3/5 | 2/5 | -3/5 | 6/5 |

**Wind (density $\propto r^{-2}$)**

| Phase | u | r | B | $N_p$ |
|---|---|---|---|---|
| Coasting | 0 | 1 | -1 | 1 |
| Blandford-McKee | -1/4 | 1/2 | -3/4 | 1/2 |
| Sedov-Taylor | -1/3 | 2/3 | -1 | 2/3 |

### 3.2 Forward Characteristic Frequencies

The synchrotron spectrum is characterized by break frequencies that evolve with time: $\nu \propto t^{\alpha}$.

**ISM**

| Phase | $\nu_m$ | $\nu_c$ | $\nu_M$ |
|---|---|---|---|
| Coasting | 0 | -2 | 0 |
| Blandford-McKee | -3/2 | -1/2 | -3/8 |
| Sedov-Taylor | -3/5 | -1/5 | 0 |

**Wind**

| Phase | $\nu_m$ | $\nu_c$ | $\nu_M$ |
|---|---|---|---|
| Coasting | -1 | -1 | 0 |
| Blandford-McKee | -3/2 | 1/2 | -1/4 |
| Sedov-Taylor | -1 | 1 | 0 |

## 3.3 Reverse Shock Dynamics (Thin Shell)

**ISM**

| Phase | u | r | B | $N_p$ |
|---|---|---|---|---|
| Crossing | 3/2 | 1 | 0 | 3/2 |
| Post-crossing | -1/4 | 1/4 | — | 0 |
| Sedov-Taylor | -2/5 | 2/5 | — | 0 |

**Wind**

| Phase | u | r | B | $N_p$ |
|---|---|---|---|---|
| Crossing | 1/2 | 1 | -1 | 1/2 |
| Post-crossing | — | 1/2 | — | 0 |
| Sedov-Taylor | — | 2/3 | — | 0 |

"—" indicates quantities not tested (insufficiently clean power-law for reliable fitting).

## 3.4 Reverse Shock Dynamics (Thick Shell)

**ISM**

| Phase | u | r | B | $N_p$ |
|---|---|---|---|---|
| Crossing | 1/4 | 1/2 | -1/4 | 1 |
| Post-crossing | -1/4 | 1/4 | — | 0 |
| Sedov-Taylor | -2/5 | 2/5 | — | 0 |

**Wind**

| Phase | u | r | B | $N_p$ |
|---|---|---|---|---|
| Crossing | 0 | 1 | -1 | 1 |
| Post-crossing | — | 1/2 | — | 0 |

| Phase | u | r | B | $N_p$ |
|---|---|---|---|---|
| Sedov-Taylor | — | 2/3 | — | 0 |

### 3.5 Reverse Shock Characteristic Frequencies

Frequency scaling is validated during the crossing phase. Post-crossing and Sedov-Taylor frequency tests are not performed for reverse shocks since the reverse shock material is no longer freshly shocked.

**Thin Shell — Crossing Phase**

| Medium | $\nu_m$ | $\nu_c$ | $\nu_M$ |
|---|---|---|---|
| ISM | 0 | -2 | 0 |
| Wind | -1 | 1 | 0 |

**Thick Shell — Crossing Phase**

| Medium | $\nu_m$ | $\nu_c$ | $\nu_M$ |
|---|---|---|---|
| ISM | — | -1 | -1/4 |
| Wind | -1 | 1 | 0 |

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## 4. Spectral Regimes

The synchrotron spectrum consists of power-law segments joined at break frequencies. The spectral index $\beta$= d(logF_ν)/d(logν) depends on the ordering of $\nu_a$, $\nu_m$, and $\nu_c$.

### Regime I: $\nu_a < \nu_m < \nu_c$ (Slow Cooling)

The standard slow-cooling spectrum where electrons cool on timescales longer than the dynamical time.

| Frequency Range | $\beta$ |
|---|---|
| $\nu < \nu_a$ | 2 |
| $\nu_a < \nu < \nu_m$ | 1/3 |
| $\nu_m < \nu < \nu_c$ | -(p-1)/2 |
| $\nu > \nu_c$ | -p/2 |

### Regime II: $\nu_m < \nu_a < \nu_c$

Self-absorption frequency lies between the injection and cooling frequencies.

| Frequency Range | $\beta$ |
|---|---|
| $\nu < \nu_m$ | 2 |

| Frequency Range | β |
|---|---|
| $\nu_m < \nu < \nu_a$ | 5/2 |
| $\nu_a < \nu < \nu_c$ | -(p-1)/2 |
| $\nu > \nu_c$ | -p/2 |

### Regime III: $\nu_a < \nu_c < \nu_m$ (Fast Cooling)

Fast-cooling regime where electrons radiate most of their energy before the next dynamical time.

| Frequency Range | β |
|---|---|
| $\nu < \nu_a$ | 2 |
| $\nu_a < \nu < \nu_c$ | 1/3 |
| $\nu_c < \nu < \nu_m$ | -1/2 |
| $\nu > \nu_m$ | -p/2 |

### Regime IV: $\nu_c < \nu_a < \nu_m$

Self-absorption occurs above the cooling frequency in fast-cooling conditions.

| Frequency Range | β |
|---|---|
| $\nu < \nu_c$ | 2 |
| $\nu_c < \nu < \nu_a$ | 2 |
| $\nu_a < \nu < \nu_m$ | -1/2 |
| $\nu > \nu_m$ | -p/2 |

### Regime V: $\nu_c < \nu_m < \nu_a$

Heavily self-absorbed fast-cooling spectrum.

| Frequency Range | β |
|---|---|
| $\nu < \nu_c$ | 2 |
| $\nu_c < \nu < \nu_m$ | 2 |
| $\nu_m < \nu < \nu_a$ | 5/2 |
| $\nu > \nu_a$ | -p/2 |

The electron index p = 2.2 is used, giving -(p-1)/2 = -0.6 and -p/2 = -1.1.

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## 5. Summary Grid Interpretation

The summary page provides a compact view of all regression tests, allowing rapid identification of any deviations from expected behavior.

**Grid Layout**

- Rows: Physical quantities (u, r, B, $N_p$ for shock; $\nu_m$, $\nu_c$, $\nu_M$ for frequencies; regimes I-V for spectra)
- Columns: Medium (ISM/Wind) subdivided by phase (Coasting, Blandford-McKee, Sedov-Taylor)

**Cell Contents**

Each cell displays the comparison between simulation and theory:

- Top value: Measured power-law exponent
- Bottom value: Expected theoretical value

**Color Coding**

| Color | Status | Criterion | | |
|---|---|---|---|---|
| Green | Pass | | measured - expected | < tolerance |
| Red | Fail | | measured - expected | >= tolerance |
| Gray | N/A | Insufficient data or not applicable | | |

**Tolerances**

- Shock dynamics: 0.1
- Characteristic frequencies: 0.1
- Spectral shapes: 0.15

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## 6. Detailed Plot Interpretation

The detailed plots provide diagnostic information for understanding any discrepancies identified in the summary grid.

### Forward Shock Dynamics Plots (2x2 grid per medium)

Each panel shows two subplots:

- Upper: Quantity vs time (log-log), with phase regions color-coded
- Lower: Local power-law exponent d(logQ)/d(logt)

Dashed lines indicate expected scaling. Markers show fitted values. Agreement between markers and dashed lines confirms correct implementation.

### Reverse Shock Dynamics Plots

Same layout as forward shock, with separate pages for thin-shell and thick-shell configurations. The crossing phase is highlighted distinctly from the post-crossing and deep Newtonian phases.

## Characteristic Frequency Plots

Same layout as shock dynamics, tracking $\nu_m$, $\nu_c$, and $\nu_M$ evolution. The frequency ordering determines which spectral regime applies at each time.

## Spectrum Shape Plots (per regime)

Each regime has a dedicated plot showing:

- Upper: $F_\nu$ vs $\nu$ with break frequencies marked as vertical lines
- Lower: Spectral index $\beta = d(\log F_\nu)/d(\log \nu)$

Colored regions indicate frequency segments between breaks. Flat regions in the lower panel confirm correct power-law behavior within each segment.

# Dynamics Summary: 60/60 passed (100%)

## Forward Shock (24/24)

| | ISM\|Coast | ISM\|Bland | ISM\|Sedov | Wind\|Coast | Wind\|Bland | Wind\|Sedov |
|---|---|---|---|---|---|---|
| $\Gamma\beta$ | -0.00<br>0 | -0.38<br>-3/8 | -0.60<br>-3/5 | -0.00<br>0 | -0.25<br>-1/4 | -0.33<br>-1/3 |
| $r$ | 1.00<br>1 | 0.27<br>1/4 | 0.40<br>2/5 | 1.00<br>1 | 0.52<br>1/2 | 0.66<br>2/3 |
| $B$ | 0.00<br>0 | -0.38<br>-3/8 | -0.60<br>-3/5 | -1.00<br>-1 | -0.78<br>-3/4 | -0.99<br>-1 |
| $N_p$ | 3.00<br>3 | 0.81<br>3/4 | 1.20<br>6/5 | 1.00<br>1 | 0.52<br>1/2 | 0.66<br>2/3 |

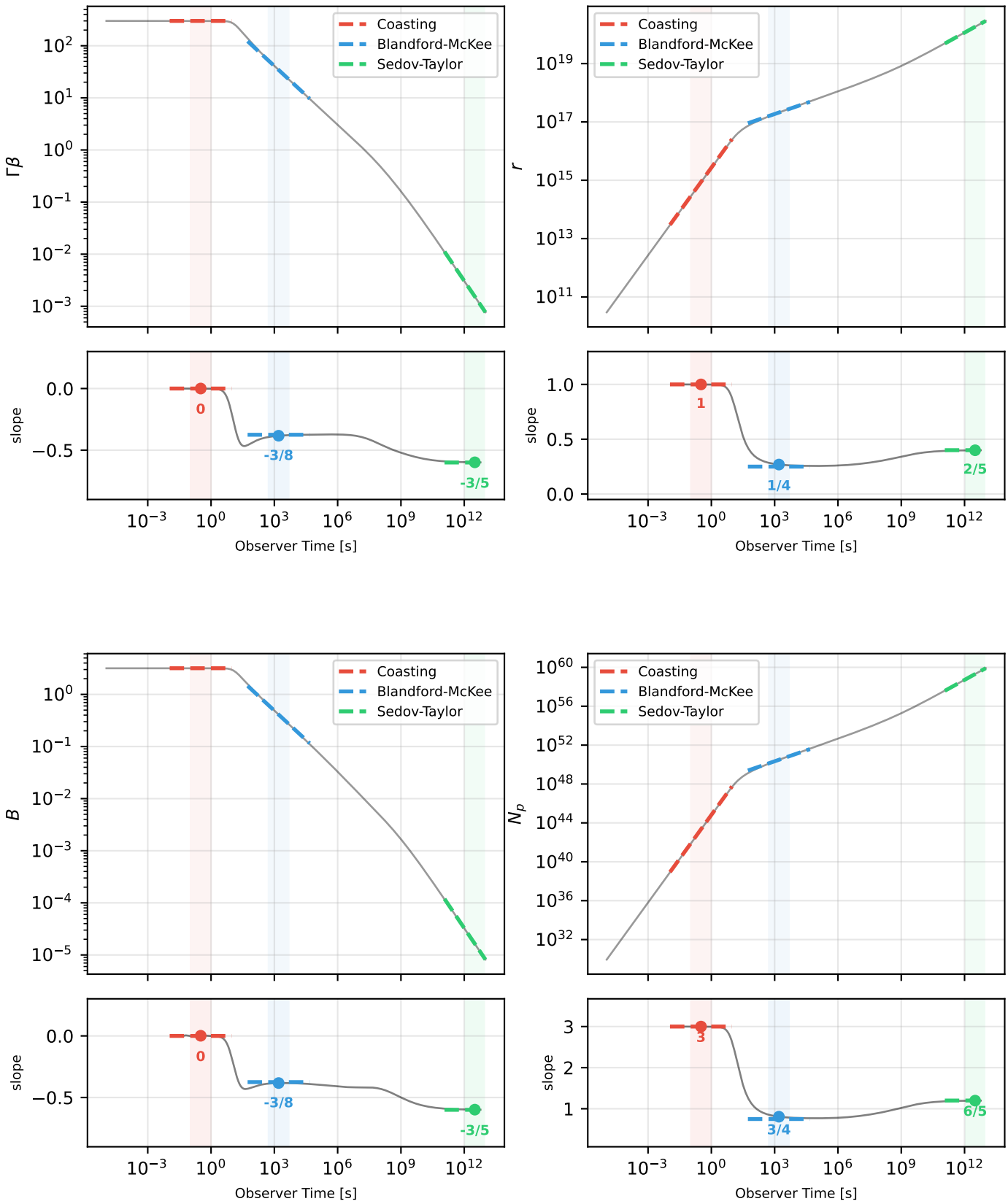
## Reverse Shock — Thin Shell (18/18)

| | ISM\|Cross | ISM\|Post- | ISM\|Sedov | Wind\|Cross | Wind\|Post- | Wind\|Sedov |
|---|---|---|---|---|---|---|
| $\Gamma\beta$ | 1.52<br>3/2 | -0.20<br>-1/4 | -0.30<br>-2/5 | 0.45<br>1/2 | — | — |
| $r$ | 1.00<br>1 | 0.26<br>1/4 | 0.40<br>2/5 | 0.97<br>1 | 0.52<br>1/2 | 0.66<br>2/3 |
| $B$ | 0.01<br>0 | — | — | -1.01<br>-1 | — | — |
| $N_p$ | 1.49<br>3/2 | 0.00<br>0 | 0.00<br>0 | 0.48<br>1/2 | 0.00<br>0 | 0.00<br>0 |

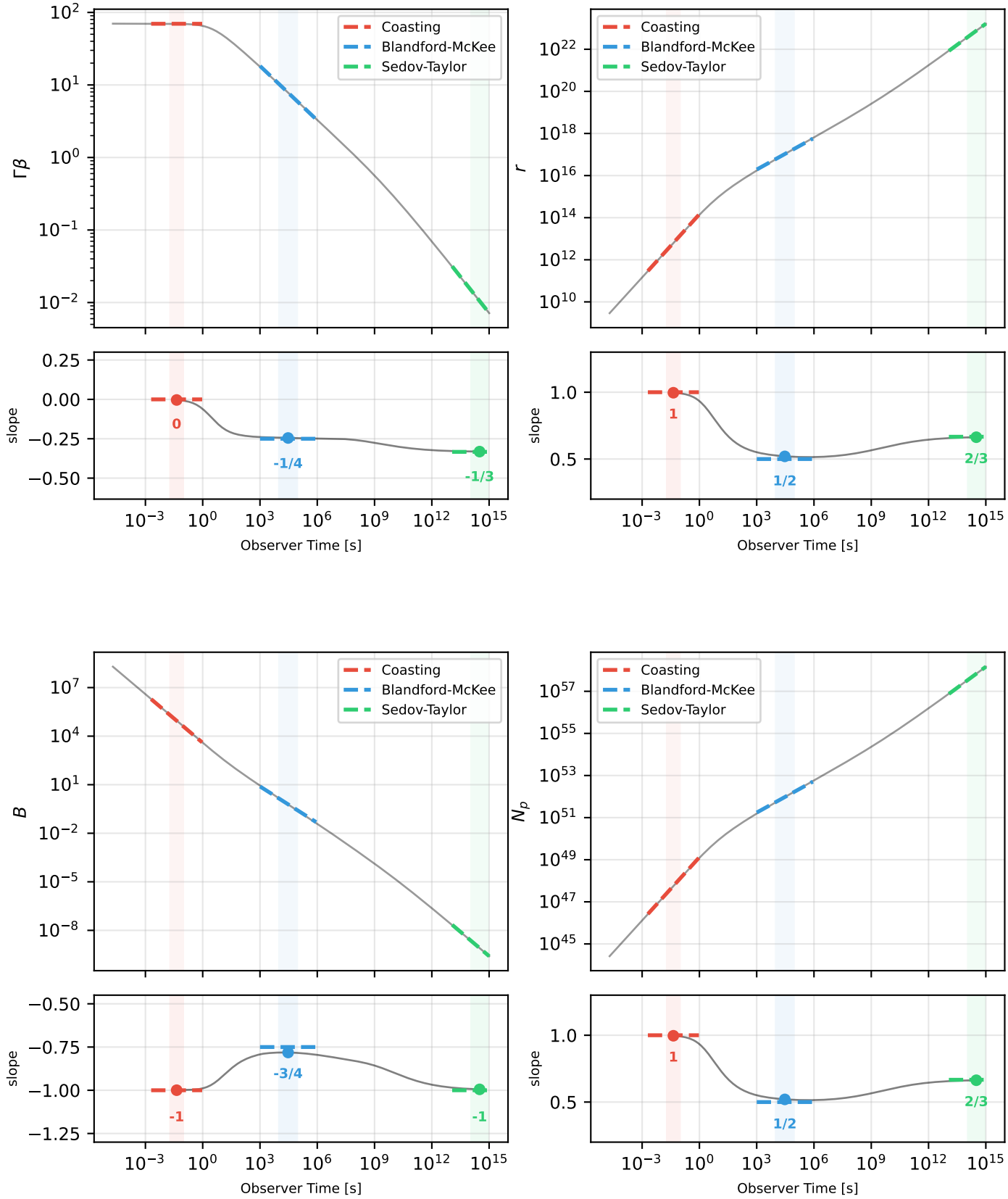
## Reverse Shock — Thick Shell (18/18)

| | ISM\|Cross | ISM\|Post- | ISM\|Sedov | Wind\|Cross | Wind\|Post- | Wind\|Sedov |
|---|---|---|---|---|---|---|
| $\Gamma\beta$ | 0.25<br>1/4 | -0.28<br>-1/4 | -0.30<br>-2/5 | -0.00<br>0 | — | — |
| $r$ | 0.50<br>1/2 | 0.30<br>1/4 | 0.40<br>2/5 | 1.00<br>1 | 0.52<br>1/2 | 0.66<br>2/3 |
| $B$ | -0.23<br>-1/4 | — | — | -1.00<br>-1 | — | — |
| $N_p$ | 1.00<br>1 | 0.00<br>0 | 0.00<br>0 | 1.00<br>1 | 0.00<br>0 | 0.00<br>0 |

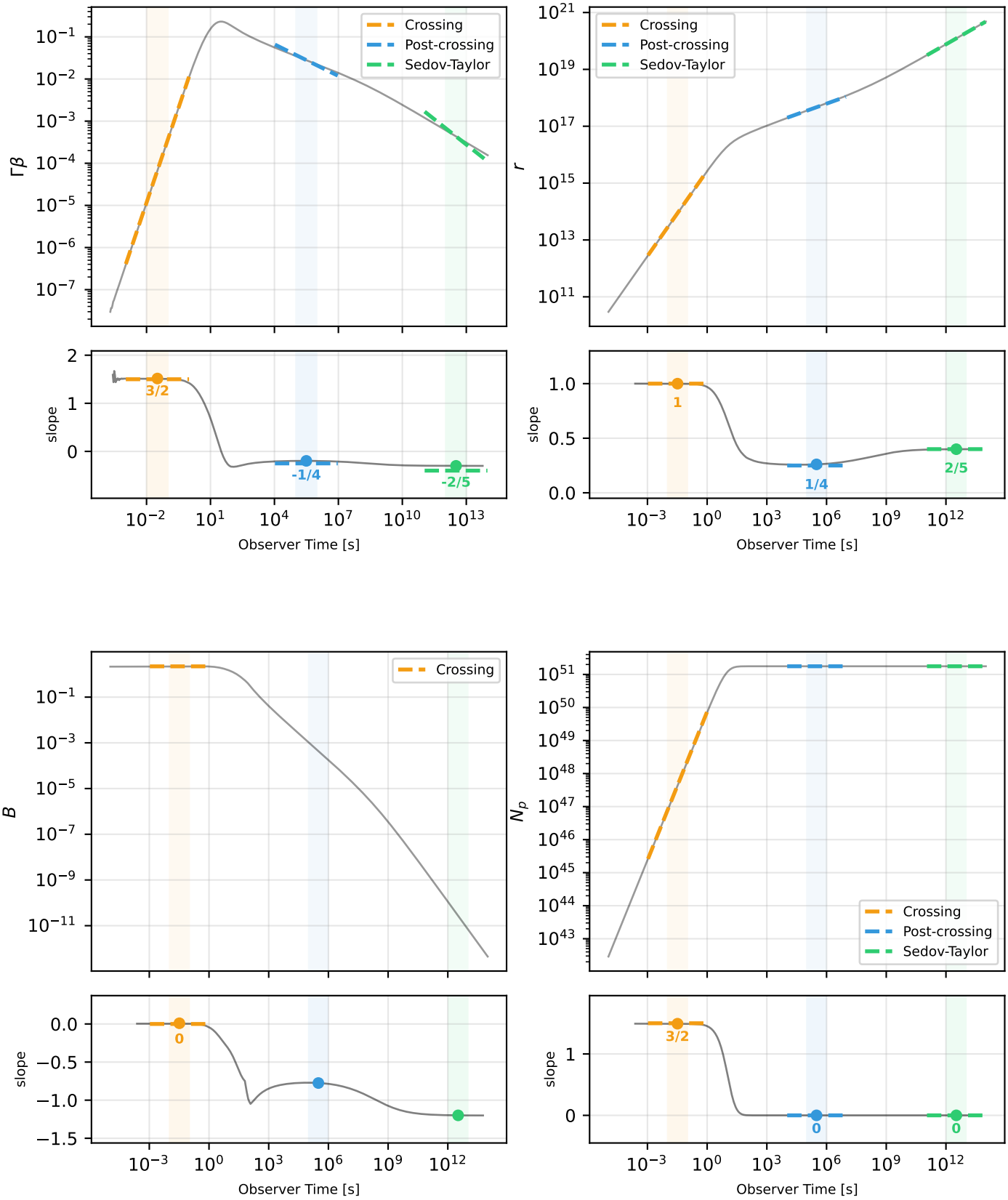
# Forward Shock Dynamics: ISM



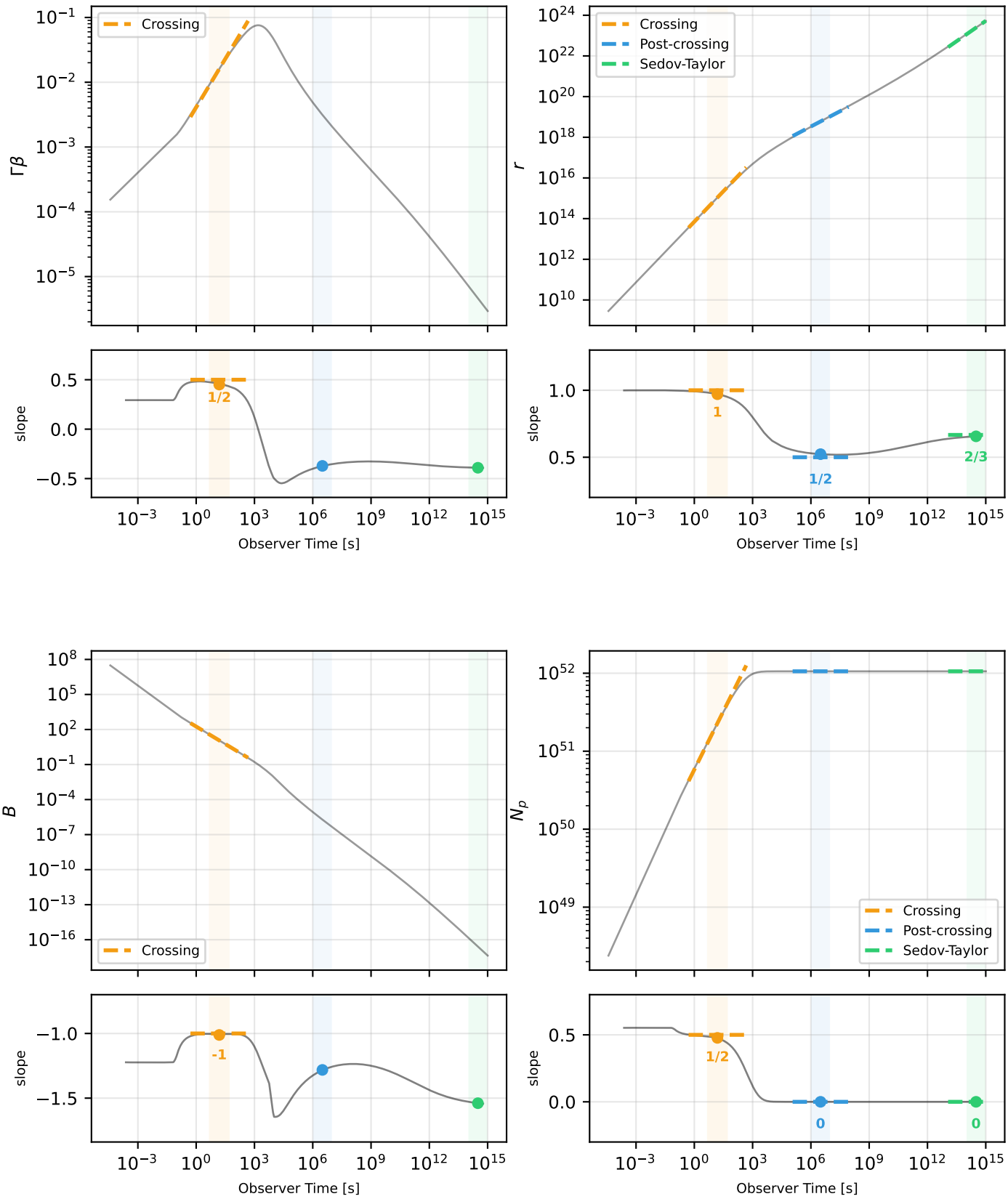
# Forward Shock Dynamics: Wind



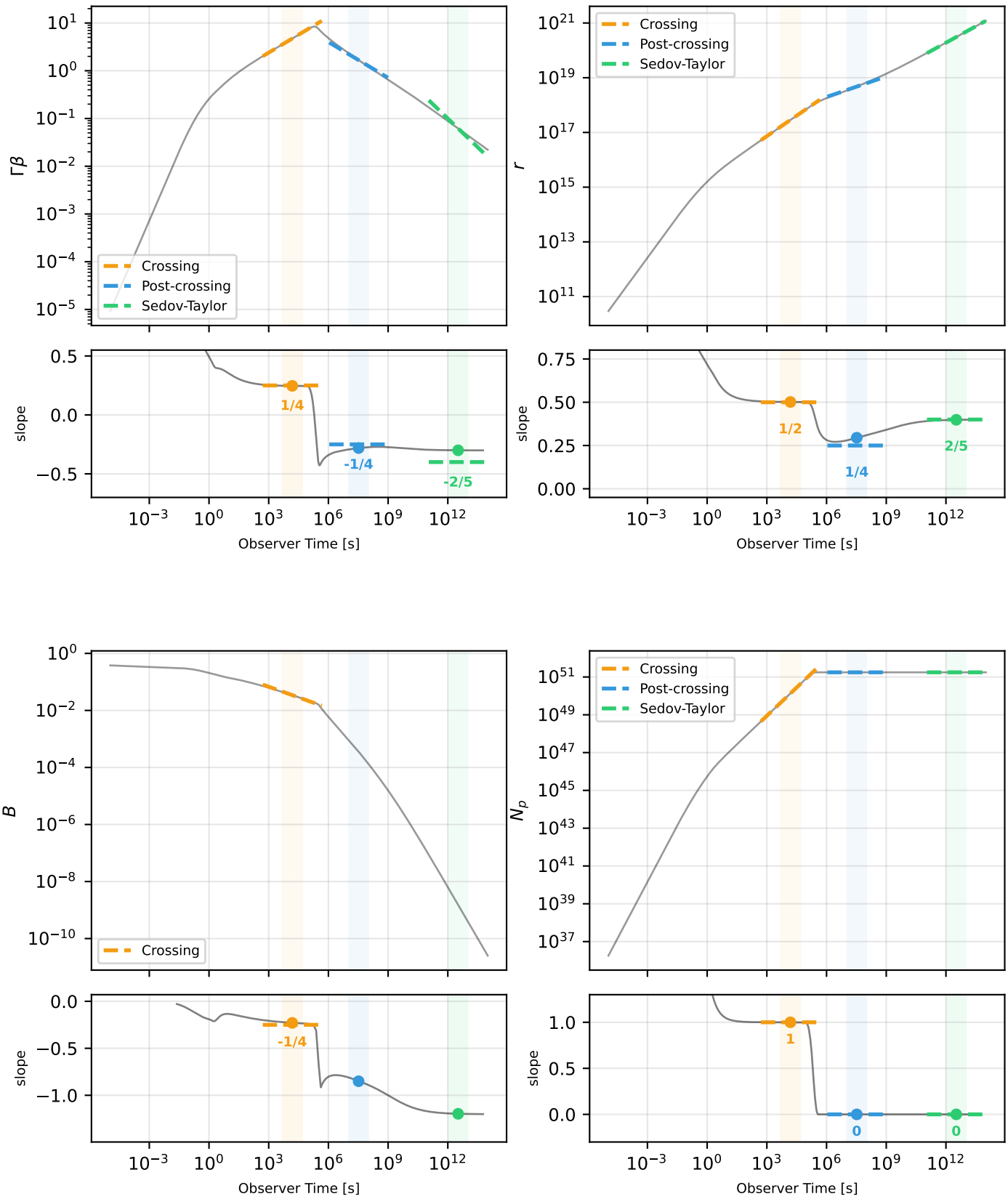
# Reverse Shock Dynamics — Thin Shell: ISM



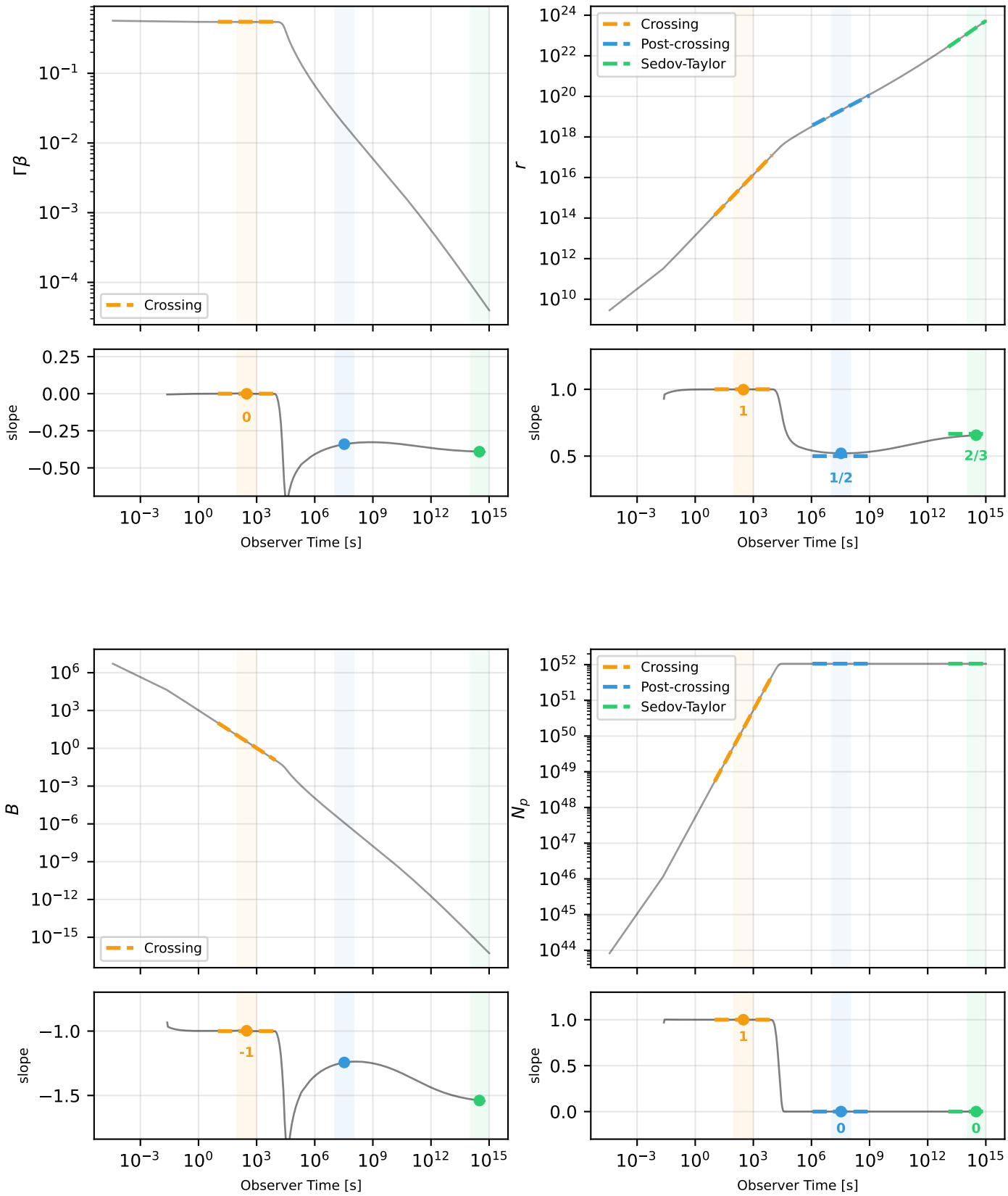
# Reverse Shock Dynamics — Thin Shell: Wind



# Reverse Shock Dynamics — Thick Shell: ISM



# Reverse Shock Dynamics — Thick Shell: Wind

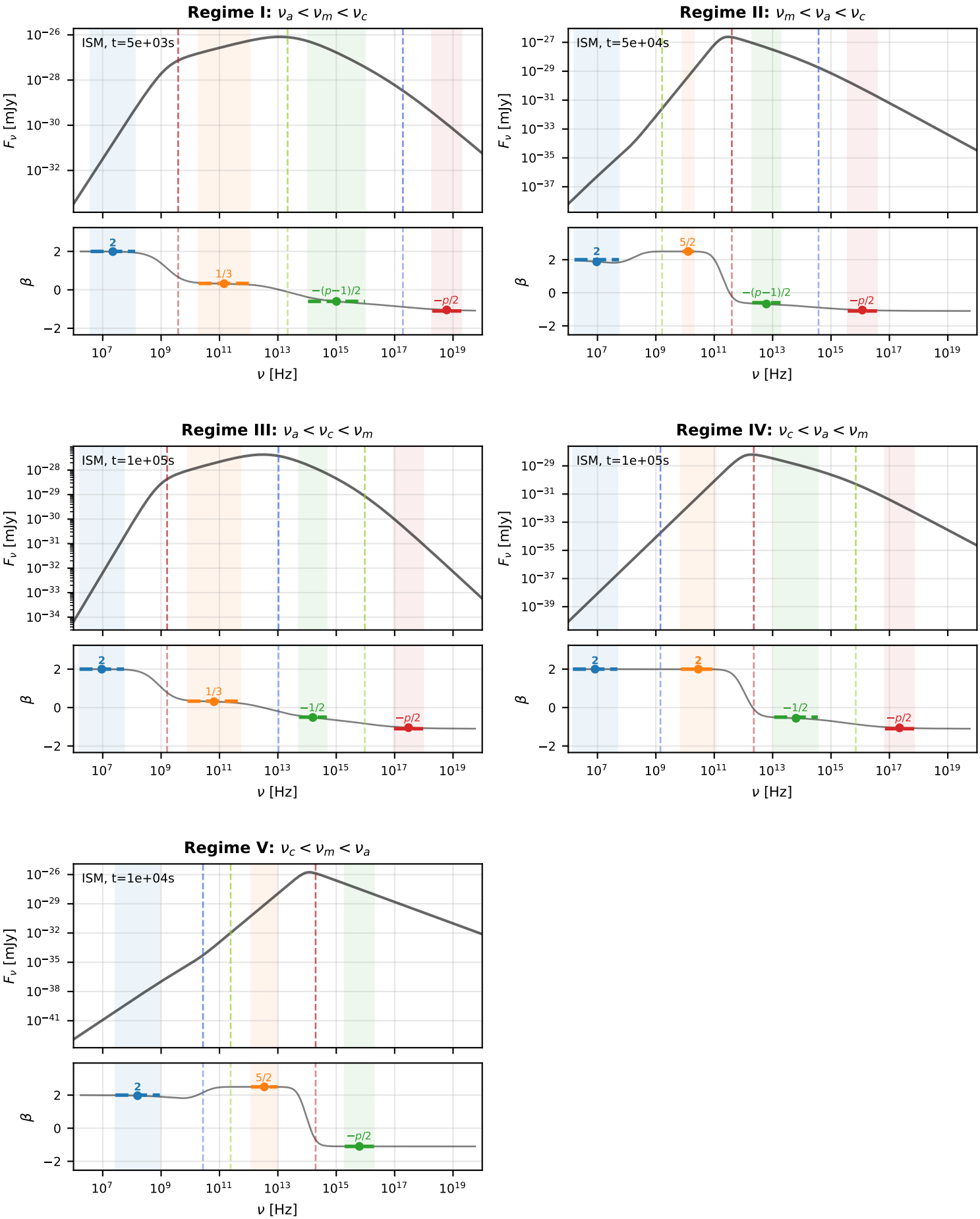


# Synchrotron Spectrum Shapes: 19/19 passed (100%)

## Synchrotron Spectrum Shapes (19/19)

| | | | | |
|---|---|---|---|---|
| | $\nu < \nu_a$ | $\nu_a < \nu < \nu_m$ | $\nu_m < \nu < \nu_c$ | $\nu > \nu_c$ |
| **Regime I** | 1.99<br>2 | 0.32<br>1/3 | -0.60<br>-(p-1)/2 | -1.04<br>-p/2 |
| | $\nu < \nu_m$ | $\nu_m < \nu < \nu_a$ | $\nu_a < \nu < \nu_c$ | $\nu > \nu_c$ |
| **Regime II** | 1.87<br>2 | 2.50<br>5/2 | -0.68<br>-(p-1)/2 | -1.04<br>-p/2 |
| | $\nu < \nu_a$ | $\nu_a < \nu < \nu_c$ | $\nu_c < \nu < \nu_m$ | $\nu > \nu_m$ |
| **Regime III** | 2.00<br>2 | 0.31<br>1/3 | -0.52<br>-1/2 | -1.05<br>-p/2 |
| | $\nu < \nu_c$ | $\nu_c < \nu < \nu_a$ | $\nu_a < \nu < \nu_m$ | $\nu > \nu_m$ |
| **Regime IV** | 2.00<br>2 | 2.00<br>2 | -0.56<br>-1/2 | -1.06<br>-p/2 |
| | $\nu < \nu_c$ | $\nu_c < \nu < \nu_m$ | $\nu_m < \nu < \nu_a$ | $\nu > \nu_a$ |
| **Regime V** | 1.97<br>2 | – | 2.50<br>5/2 | -1.10<br>-p/2 |

# Synchrotron Spectrum Shapes: Spectral Index Verification



# Frequencies Summary: 29/29 passed (100%)

## Forward Shock (18/18)

| | ISM\|Coast | ISM\|Bland | ISM\|Sedov | Wind\|Coast | Wind\|Bland | Wind\|Sedov |
|---|---|---|---|---|---|---|
| $\nu_m$ | 0.01<br>0 | -1.52<br>-3/2 | -0.60<br>-3/5 | -1.01<br>-1 | -1.54<br>-3/2 | -1.00<br>-1 |
| $\nu_c$ | -2.00<br>-2 | -0.46<br>-1/2 | -0.20<br>-1/5 | -0.97<br>-1 | 0.58<br>1/2 | 0.99<br>1 |
| $\nu_M$ | -0.00<br>0 | -0.38<br>-3/8 | -0.00<br>0 | -0.00<br>0 | -0.24<br>-1/4 | -0.00<br>0 |

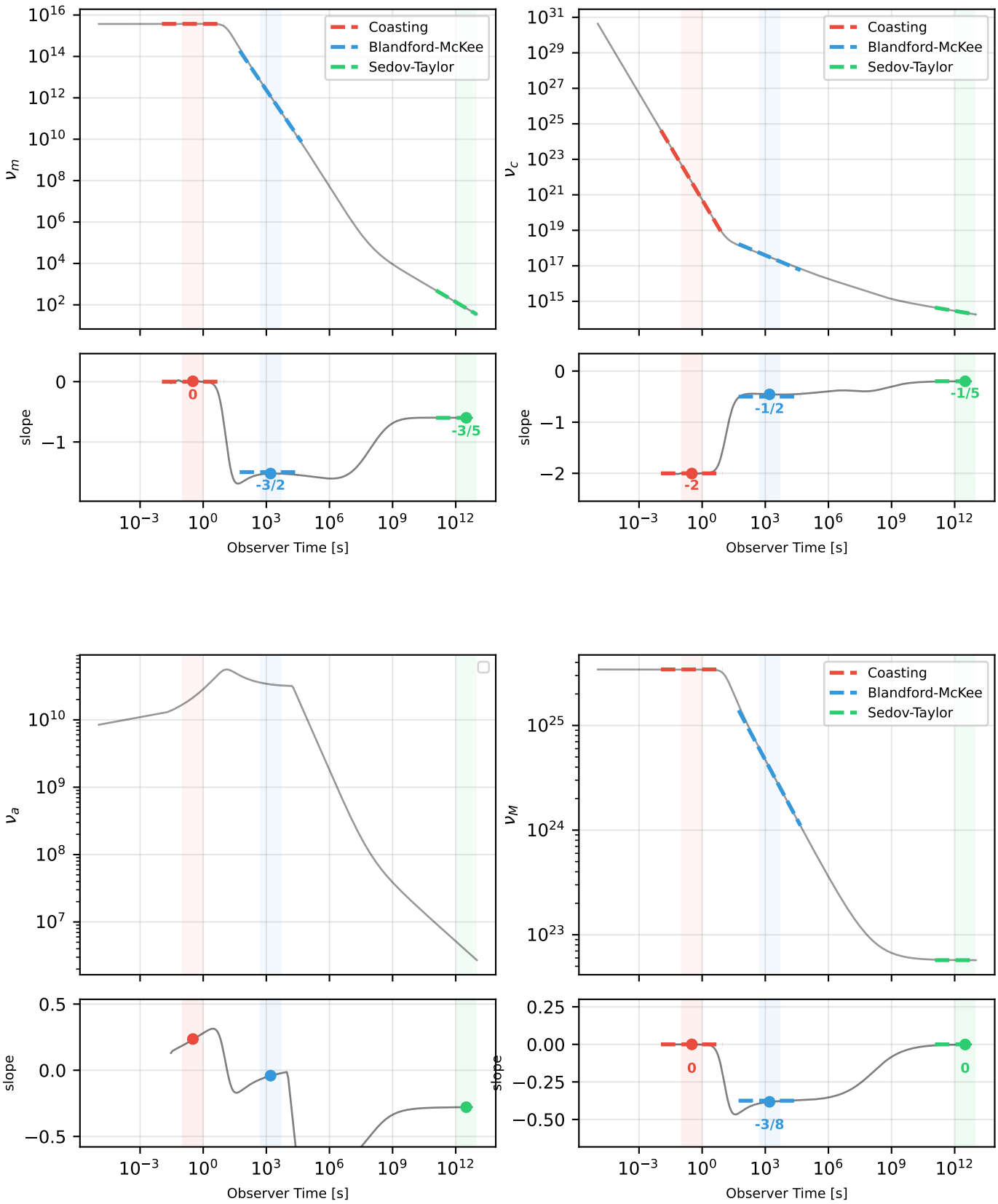
## Reverse Shock — Thin Shell (6/6)

| | ISM\|Cross | ISM\|Post- | ISM\|Sedov | Wind\|Cross | Wind\|Post- | Wind\|Sedov |
|---|---|---|---|---|---|---|
| $\nu_m$ | 0.01<br>0 | — | — | -1.03<br>-1 | — | — |
| $\nu_c$ | -2.03<br>-2 | — | — | 1.04<br>1 | — | — |
| $\nu_M$ | -0.00<br>0 | — | — | -0.02<br>0 | — | — |

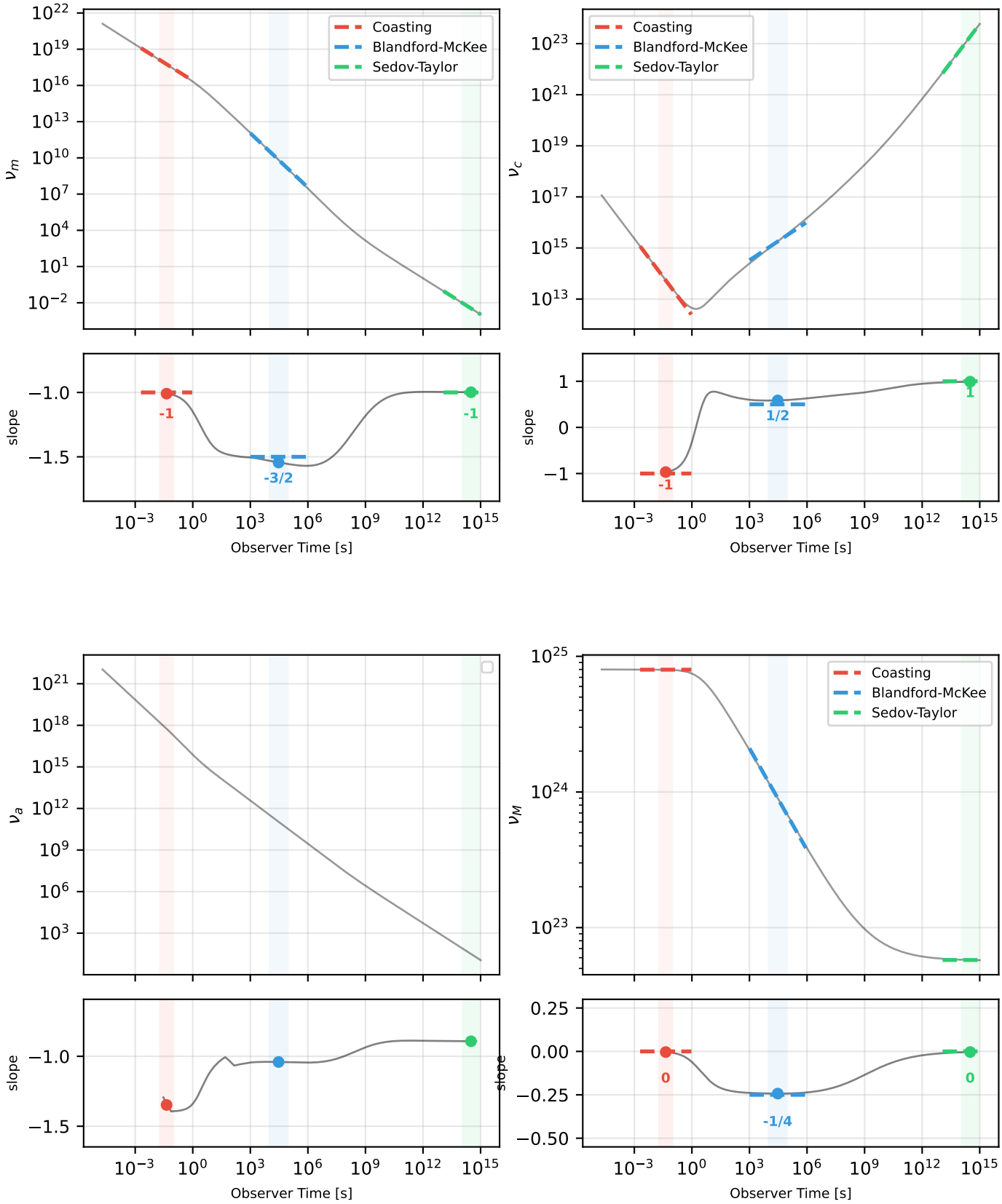
## Reverse Shock — Thick Shell (5/5)

| | ISM\|Cross | ISM\|Post- | ISM\|Sedov | Wind\|Cross | Wind\|Post- | Wind\|Sedov |
|---|---|---|---|---|---|---|
| $\nu_m$ | — | — | — | -1.00<br>-1 | — | — |
| $\nu_c$ | -1.06<br>-1 | — | — | 1.00<br>1 | — | — |
| $\nu_M$ | -0.25<br>-1/4 | — | — | 0.00<br>0 | — | — |

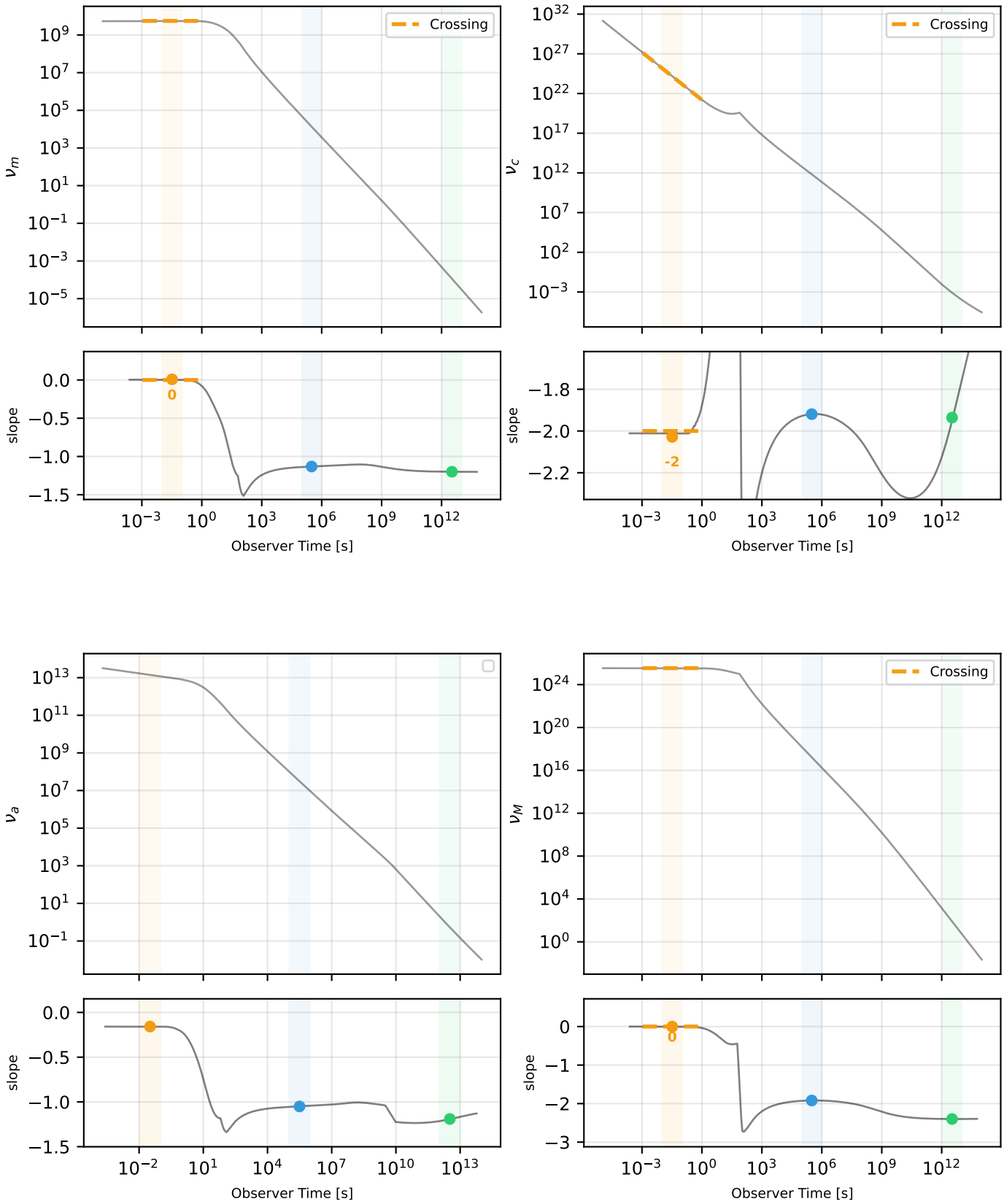
# Forward Shock Frequencies: ISM



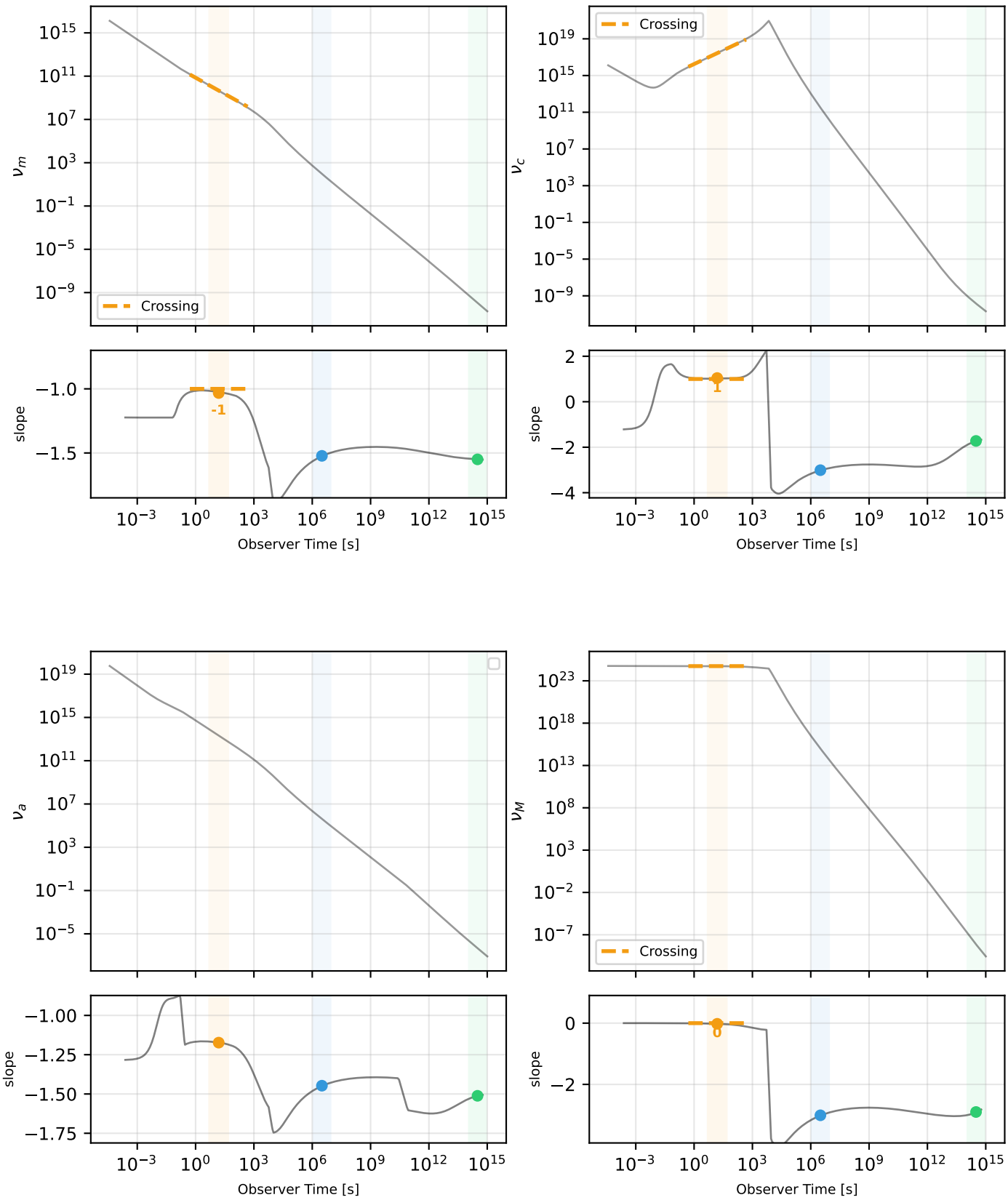
# Forward Shock Frequencies: Wind



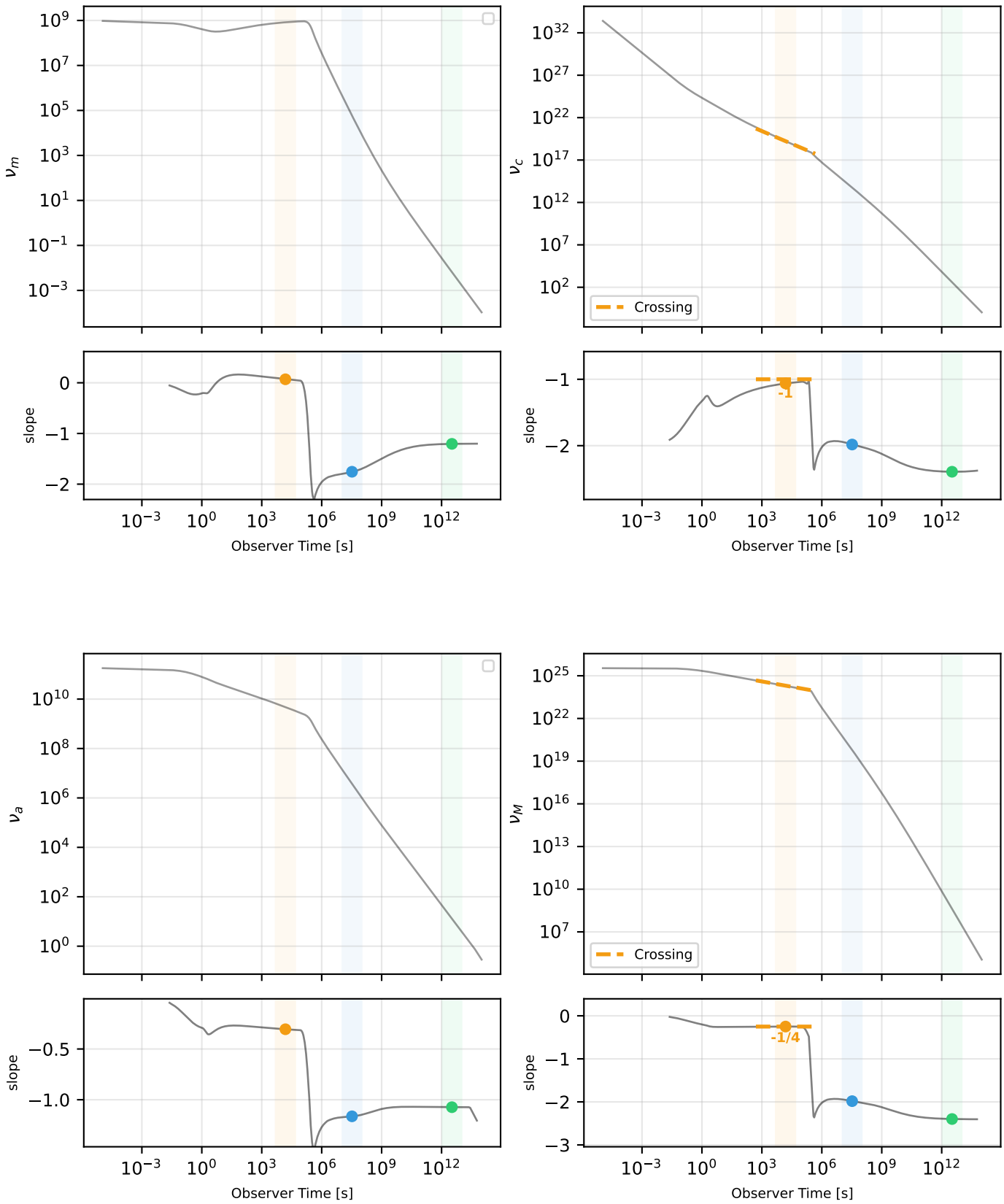
# Reverse Shock Frequencies — Thin Shell: ISM



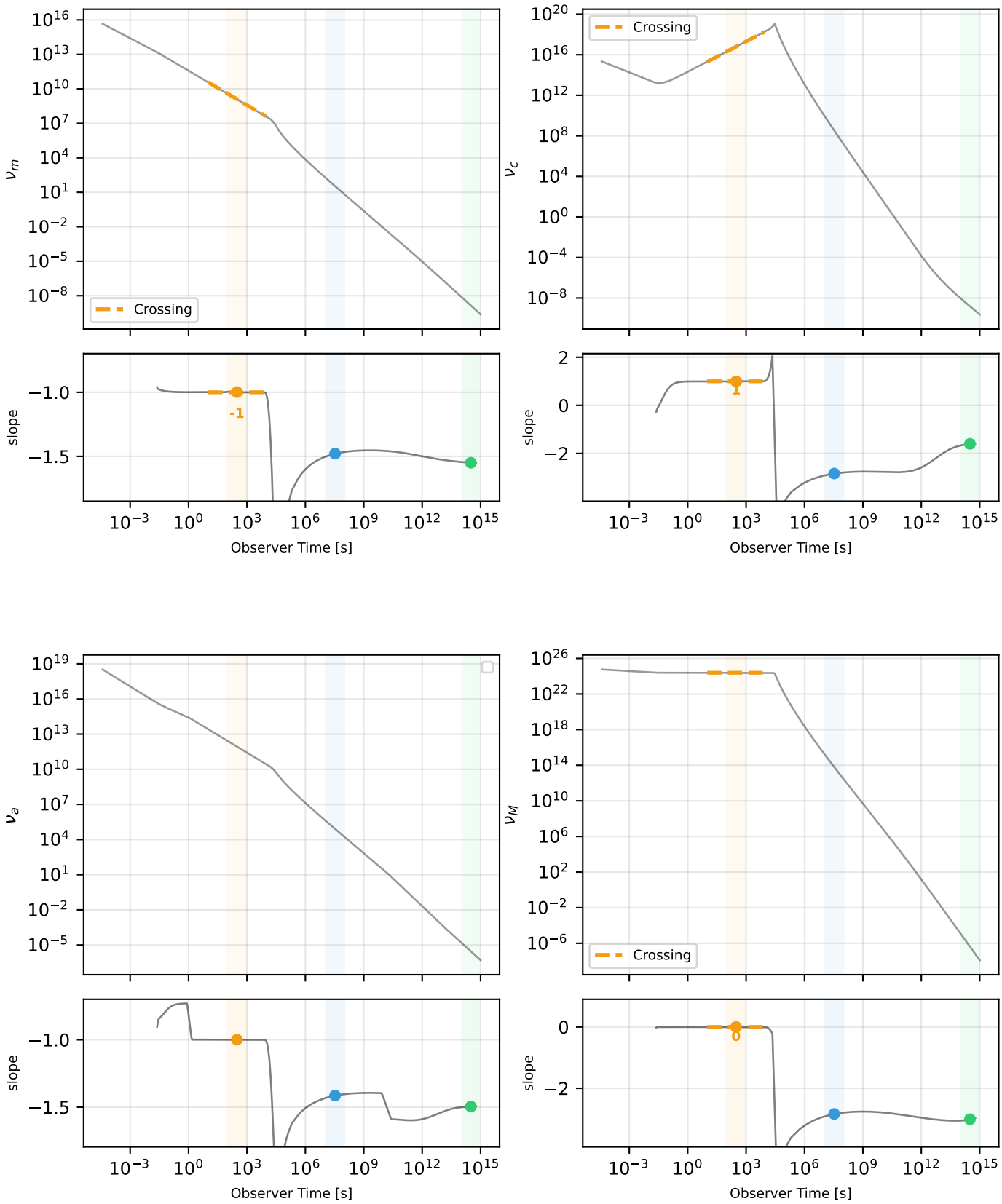
# Reverse Shock Frequencies — Thin Shell: Wind



# Reverse Shock Frequencies — Thick Shell: ISM



# Reverse Shock Frequencies — Thick Shell: Wind



Section 2

# Benchmark Results

Performance timing and resolution convergence analysis

# Benchmark Report Guide

This document describes the benchmark validation framework for VegasAfterglow, which assesses numerical convergence and computational performance across physical configurations.

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## 1. Configuration Space

The benchmark suite systematically tests a grid of physical scenarios and numerical settings to characterize code behavior across the parameter space relevant for GRB afterglow modeling.

### 1.1 Physical Parameters

| Parameter | Options | Description |
|---|---|---|
| Jet Structure | tophat, gaussian, powerlaw, two_component | Angular energy profile |
| External Medium | ISM, Wind | Density profile: constant (ISM) or $\propto r^{-2}$ (Wind) |
| Radiation | synchrotron, full_ssc, fast_cooling, steep/flat_spectrum, rvs_sync_thin, rvs_sync_thick | Radiation physics |
| Viewing Angle | $\theta_v/\theta_c$ = 0, 2, 4 | On-axis (0) vs off-axis (>1) |

### 1.2 Numerical Resolution

Resolution controls the discretization density in each computational dimension. The fiducial values represent the recommended default settings. Note that θ and t grids enforce a minimum total point count regardless of the ppd value, so low ppd settings may not reduce the actual grid size.

| Dimension | Symbol | Unit | Fiducial | Test Range |
|---|---|---|---|---|
| Azimuthal angle | φ | per degree | 0.1 | 0.1 - 0.25 |
| Polar angle | θ | per degree | 0.25 | 0.25 - 2.0 |
| Observer time | t | per decade | 10 | 10 - 25 |

### 1.3 Frequency Bands

Convergence is evaluated independently at three representative frequencies spanning the electromagnetic spectrum.

| Band | Frequency (Hz) |
|---|---|
| Radio | $10^9$ |

| Band | Frequency (Hz) |
|---|---|
| Optical | $4.84 \times 10^{14}$ |
| X-ray | $10^{18}$ |

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## 2. Convergence Criteria

Numerical accuracy is quantified by comparing results at each resolution to a high-resolution reference run. Both mean and maximum relative errors across the light curve are evaluated.

| Status | Criteria |
|---|---|
| PASS | mean error < 5% AND max error < 15% |
| ACCEPTABLE | mean error < 5% AND max error >= 15% |
| FAIL | mean error >= 5% |

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## 3. Summary Grid

The summary grid provides a compact overview of convergence status for all tested configurations on a single page.

### 3.1 Layout

Each cell represents one model configuration (jet, medium, viewing angle). Cells are arranged to facilitate comparison across jet types and viewing angles.

### 3.2 Cell Contents

| Line | Content |
|---|---|
| 1 | Model ID |
| 2 | Configuration shorthand (jet/medium/angle_ratio) |
| 3 | Maximum error at fiducial resolution |

### 3.3 Color Coding

| Color | Status |
|---|---|
| Green | PASS |
| Blue | ACCEPTABLE |
| Pink | FAIL |

| Color | Status |
|---|---|
| Gray | No data |

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## 4. Overview Plots

The overview page provides performance profiling across configurations, helping identify computational bottlenecks and compare execution times.

### 4.1 Panel Layout

| Position | Content |
|---|---|
| Top-left | Light curve computation time by jet type |
| Top-right | Stage breakdown (profiling build) or resolution cost scaling |
| Bottom-left | Medium comparison (ISM vs Wind) |
| Bottom-right | Wind/ISM speed ratio |

### 4.2 Timing Metric

Each configuration is timed by computing a 30-point broadband light curve ($t = 10^2$ to $10^8$ s) at the fiducial resolution. The reported time includes dynamics computation and flux evaluation in a single `flux_density` call.

### 4.3 Stage Breakdown

When built with profiling enabled (`pip install -e ".[test]" --config-settings=cmake.define.AFTERGLOW_PROFILE=ON`), the top-right panel shows a stacked bar chart of per-stage CPU cost for each jet/medium combination. The stages correspond to the internal C++ computation pipeline:

| Stage | Description |
|---|---|
| mesh | Coordinate grid generation |
| shock_dynamics | Forward/reverse shock ODE integration |
| EAT_grid | Equal arrival time surface grid |
| syn_electrons | Synchrotron electron distribution |
| syn_photons | Synchrotron photon spectrum |
| cooling | SSC/IC cooling corrections |
| sync_flux | Synchrotron flux integration |
| ic_photons | Inverse Compton photon spectrum |

| Stage | Description |
|---|---|
| ssc_flux | SSC flux integration |

Without profiling, the panel falls back to showing total resolution cost scaling.

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## 5. Per-Model Convergence Pages

Each configuration receives a detailed convergence analysis page showing how accuracy and performance vary with resolution.

### 5.1 Row Contents

The page displays a 4x3 grid where each column corresponds to one resolution dimension (φ, θ, t) and each row shows a different metric.

| Row | Y-axis | Threshold |
|---|---|---|
| 1 | Maximum relative error | 15% |
| 2 | Mean relative error | 5% |
| 3 | CPU time (ms) | - |
| 4 | Flux (mJy) | - |

### 5.2 Plot Features

| Feature | Meaning |
|---|---|
| Star marker | Fiducial resolution |
| Dashed line | Error threshold |
| Solid curves | Resolution >= fiducial |
| Dotted curves | Resolution < fiducial |

### 5.3 Status Indicator

The page header displays convergence status with color coding matching Section 3.3.

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## 6. Interpreting Results

### 6.1 Light Curve Convergence

The bottom row plots all tested resolutions together on the same axes. Visual spread between curves indicates resolution dependence.

| Pattern | Interpretation |
|---|---|
| No spread (curves overlap) | Converged |
| Spread in dashed lines only | Not converged below fiducial, acceptable at fiducial |
| Spread in solid lines | Not converged even above fiducial resolution |

## 6.2 Error Convergence

The top two rows show how errors decrease as resolution increases. The shape of these curves indicates convergence quality.

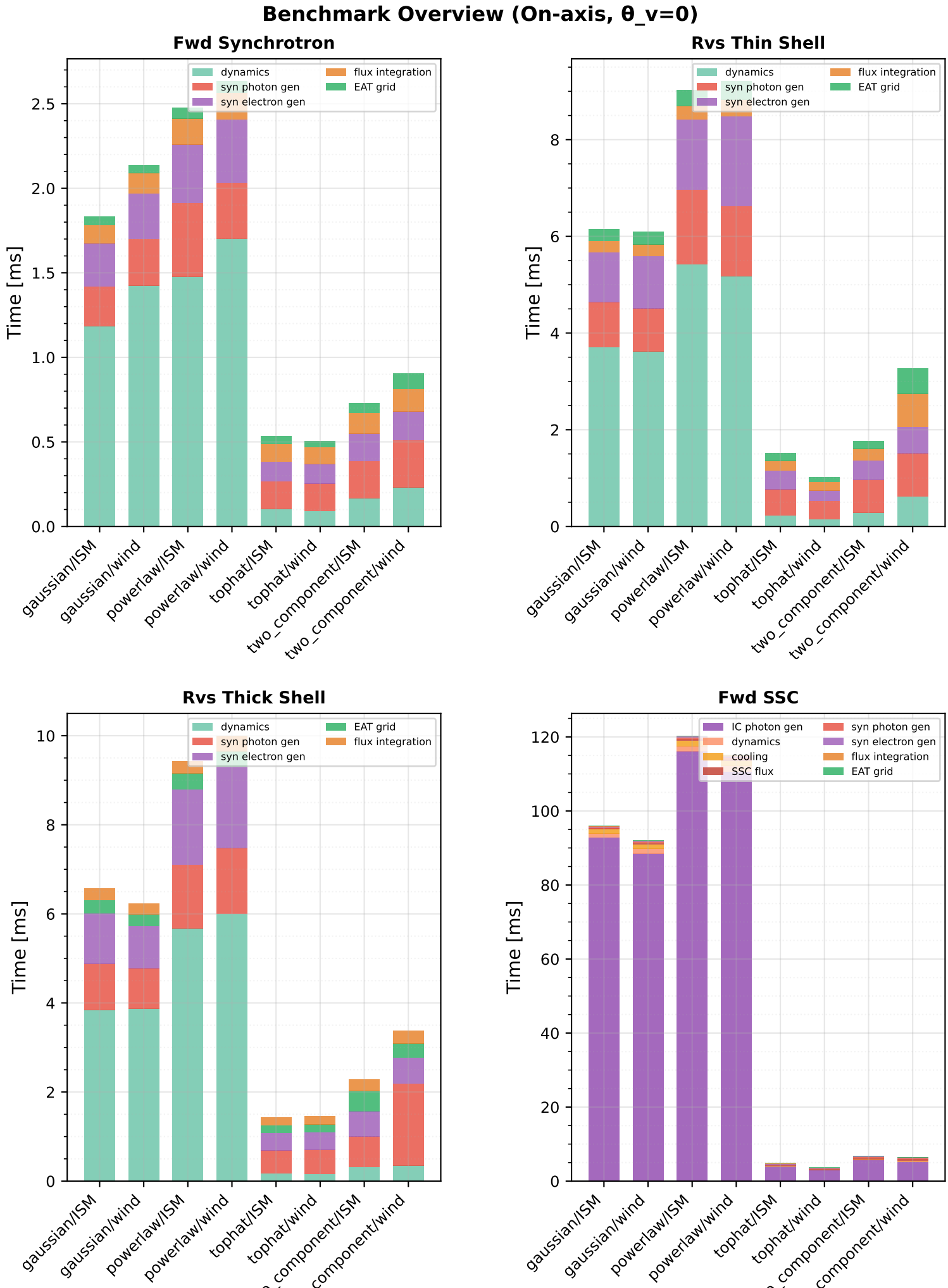
| Pattern | Interpretation |
|---|---|
| Monotonic decrease | Stable convergence |
| Plateau | Noise floor or discretization limit |
| Non-monotonic | Potential numerical instability |

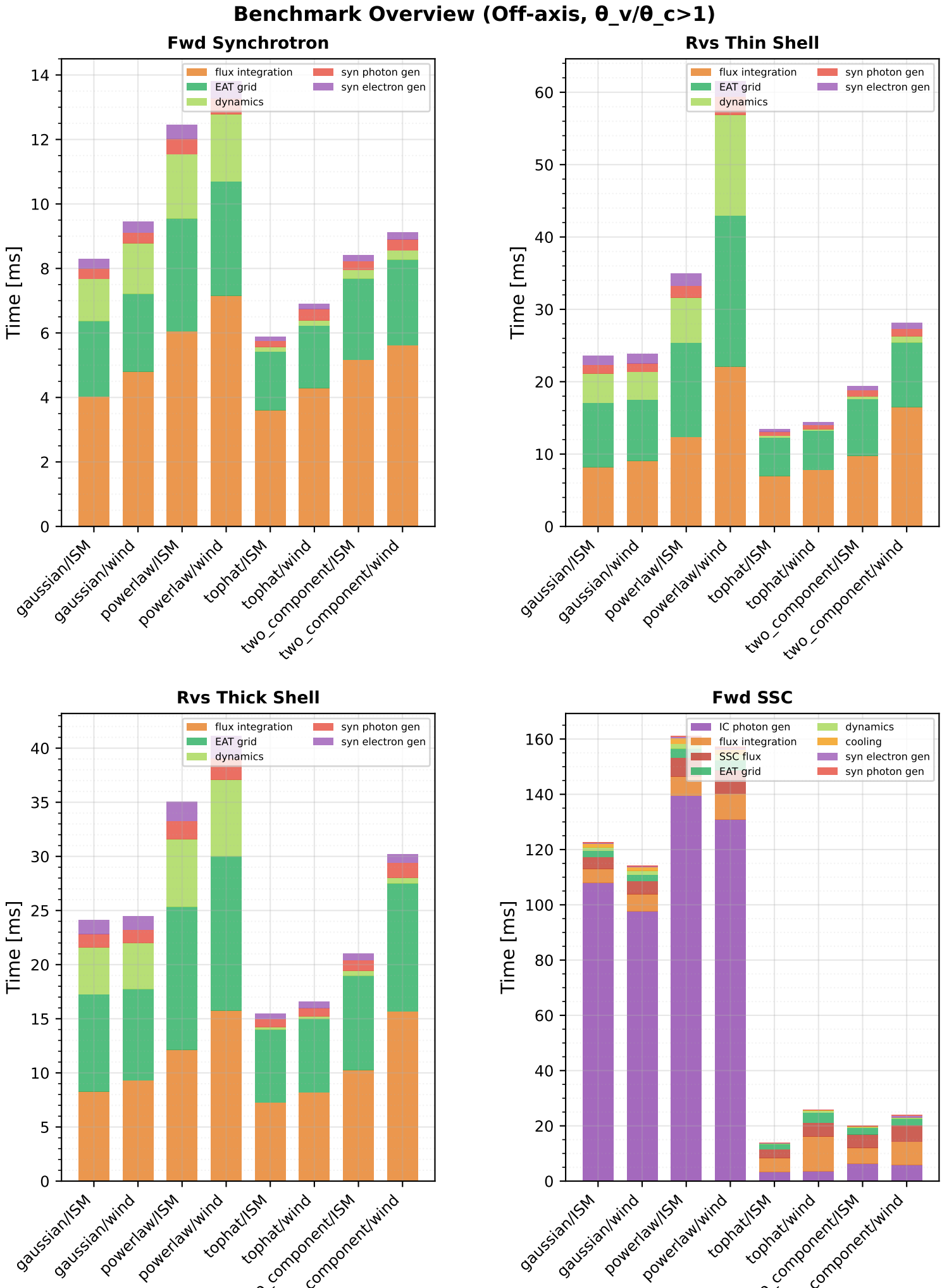
## 6.3 Performance Scaling

Computational cost typically scales as:

- Linear with time resolution ($t_p$pd)
- Quadratic with angular resolution ($\varphi_p$pd $\times\theta_p$pd)

Wind medium simulations generally require more computation due to the radially-varying density profile. However, the medium-aware adaptive grid often produces smaller grids for wind (earlier deceleration time), which can offset this cost.


Benchmark Overview (On-axis, θ_v=0)
Fwd Synchrotron
Rvs Thin Shell
Rvs Thick Shell
Fwd SSC
Time [ms]
dynamics
syn photon gen
syn electron gen
flux integration
EAT grid
IC photon gen
cooling
SSC flux
gaussian/ISM
gaussian/wind
powerlaw/ISM
powerlaw/wind
tophat/ISM
tophat/wind
two_component/ISM
two_component/wind


Benchmark Overview (Off-axis, θ_v/θ_c>1)
Fwd Synchrotron
Rvs Thin Shell
Rvs Thick Shell
Fwd SSC
Time [ms]
flux integration
EAT grid
dynamics
syn photon gen
syn electron gen
IC photon gen
SSC flux
cooling
gaussian/ISM
gaussian/wind
powerlaw/ISM
powerlaw/wind
tophat/ISM
tophat/wind
two_component/ISM
two_component/wind